

DEEPAK KUMAR

Senior AI Engineer - Agentic AI & Generative AI Systems

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EDUCATION

2019	Indian Institute of Technology, Bombay <i>Master's in Computer Science, Machine Learning</i>	GPA: 8.20/10
2017	Dr B R Ambedkar National Institute of Technology, Jalandhar <i>B.Tech in Information Technology</i>	GPA: 8.67/10

EXPERIENCE

JAN'23 - PRESENT	FOUNDINGLABS.AI INDIA <i>Senior AI Engineer</i> • Agentic AI: Architected the company's production multi-agent AI platform on LangGraph and LangChain, orchestrating planner, supervisor, reflection, and reviewer agents with human-in-the-loop checkpoints – enabling long-running, multi-step workflows to execute reliably without manual intervention. • Advanced RAG: Designed a hybrid, reasoning-based retrieval architecture – combining BM25, dense embeddings, reciprocal rank fusion, metadata filtering, reranking, knowledge graphs, and PageIndex-based document reasoning – to efficiently process hundreds of documents daily into accurate, context-aware answers. • AI Platform: Owned the design and scaling of backend AI microservices on FastAPI and async REST APIs, containerized on Docker and Kubernetes across AWS, taking the platform from prototype to around 5,000 monthly active users and 100 concurrent sessions at P95 latency under 5 seconds. • Reliability & Evaluation: Built the validation layer – Pydantic guardrails, confidence scoring, LLM-as-a-judge, and RAGAS evaluation – that every AI feature must pass before shipping, cutting the hallucination rate from an estimated 12% to under 2%, and stood up a continuous evaluation pipeline tracking retrieval precision, faithfulness, hallucination rate, latency, and cost on every release. • LLM Observability & Testing: Used LangSmith for end-to-end prompt and agent tracing, and Promptfoo for regression testing across prompt versions, catching quality drops before they reached production and giving the team full visibility into every agent's reasoning steps. • Deployment: Owned cloud infrastructure on AWS (EC2, EKS, S3, RDS, IAM, ALB, CloudWatch) with GitHub Actions CI/CD, moving the team to zero-downtime rolling releases with end-to-end observability via Prometheus and Grafana – cutting deployment incidents by roughly 40%.
JUNE'22 - DEC'22	INTEL INDIA <i>Senior Software Engineer</i> • Intel SSM Team: Developed system simulation infrastructure for firmware and hardware validation as part of Intel's System Simulation and Modelling (SSM) team. • Innovation: Developed various features (in C++ and Python) like VTIO secure USB connection, xHCI modules, xDCI modules as the owner of Security IP modules. • Resolving Bugs: Identified and fixed various code bugs, technical debts and code migration.
JULY'19 - JUNE'22	MICRON SEMICONDUCTOR SINGAPORE <i>NVE Product Engineer, Data Analyst</i> • Deep Learning: Implemented Feed Forward Neural Network(in PyTorch) to predict material's future intrinsic properties after analysing features impacting material variations. • Success Cases: Implemented 200+ success cases with high precision and low recall to screen failing materials(classify good and bad material) to meet customer requirements based on data mining including various stacked ML algorithms. • Innovation: Proposed new multivariate methods to improve metrics correlation and proposed new mathematical formulas to project future material variations based on past material performance. • Automation: Automated various existing methodologies, daily and weekly reports reducing human effort and saving time.

KEY PROJECTS

- **Personal AI Communication Agent** (June'26)
 - Fine-tuned an open-source LLM on WhatsApp conversations to learn writing style, tone, and conversational patterns for personalized response generation.
 - Built a **LangGraph-based autonomous agent** with RAG memory, enabling context-aware, human-like replies and **automated message** handling through messaging APIs.
- **Research Intelligence Assistant** (June'26)
 - Designed a research intelligence assistant using **Agentic AI** and **RAG** to analyze research papers and generate citation-backed literature reviews.
 - Implemented multi-document retrieval, **semantic search**, query decomposition, and evidence-based question answering over research papers using **vector databases**.
- **Medical Report Analysis AI Assistant** (June'26)
 - Built an Agentic AI assistant to analyze laboratory reports, medications, and clinical records using specialized agents for **report interpretation**, **risk assessment**, drug analysis, multilingual translation, and patient education.
 - Implemented a **Medical RAG** pipeline over clinical guidelines, drug databases, and healthcare literature to generate evidence-backed explanations, identify potential health risks, and **answer patient queries** in **regional languages**.

MAJOR THESIS PROJECT

JULY'18 - JUNE'19	COMPUTATIONAL ANALYSIS OF JUDICIAL DATA <i>Master's Thesis under PROF. SHIVARAM KALYANAKRISHNAN , IIT Bombay</i> • Developed a web interface for lawyers to extract structured information from judicial documents using numerical relation extraction and LDA for topic classification. • Built predictive models (LSTM, SVM, and Naive Bayes) to classify case outcomes as Allowed, Dismissed, or Partly Allowed. • Extracted disputed amounts from long judicial documents using numerical relation extraction. • Improved server response time with Celery and optimized throughput using Redis caching.
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ACADEMIC PROJECTS

- **Object Classifier** under Prof. Vinayak Naik (Jan'18 - May'18)
 - **Objective:** Developed an **Android** application to detect and identify objects in real time and used narrate categories to assist blind people.
 - Used **SSD(Single Shot multibox Detection) Mobilenet Model** for prediction compared to **Faster R-FCN** because of its lightweight architecture.
- **Software planning and resource management system** under Prof. N.L Sarda (Jan'18 - May'18)
 - **Objective :** Developed a **JavaFX** application for project planning, resource management, and execution.
 - Used **IntelliJ IDEA**, **JUnit**, **Maven**, and **GitHub** for development, testing, build management, version control, code review, and release management.
 - Prepared Software Requirements Specification (SRS) and design documentation.

TECHNICAL SKILLS

Generative AI	LangGraph, LangChain, Agentic AI, Multi-Agent Systems, RAG, Hybrid Search, MCP, Prompt Engineering, Function Calling, Tool Calling, AI Chatbots
LLMs & NLP Languages	GPT-4, Claude, Gemini, Llama, Ollama, OpenRouter, BERT, Knowledge Graphs
Backend & Cloud	Python, C++, Java, SQL
Vector Databases	FastAPI, REST APIs, Docker, Kubernetes, Redis, PostgreSQL, AWS, CI/CD, Git, GitHub
AI Observability	FAISS, ChromaDB, Pinecone, Qdrant
Machine Learning	LangSmith, Prompt Evaluation, Prompt Tracing, AI Evaluation, Hallucination Detection, Testing
Computer Vision	PyTorch, TensorFlow, Scikit-learn, NumPy, Reinforcement Learning
Analytics & Visualization	YOLO, SSD MobileNet, Faster R-CNN, MinerU
Others	Power BI, Tableau, JMP
	AI Search Engines, $\LaTeX$, DS Algorithms